

Shortest-geodesic supports in triangle-free maximum cuts: an infinite Hall obstruction and the first minimal footprints

Alper Ferudun
alper@mercurycodelab.com

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Abstract

Fix a cut of a triangle-free graph, write B for its crossing edges and M for its monochromatic edges, and associate to each $xy \in M$ the B -edges lying on shortest x - y paths. A natural local approach to triangle-free bipartization asks whether these supports satisfy Hall expansion. We show that they need not, even when the cut is the unique maximum cut and every bad edge has a unique shortest geodesic.

For every $t \geq 1$, we construct a triangle-free graph G_t on $7t + 3$ vertices with a unique maximum cut and $\text{bip}(G_t) = t^2$. Its t^2 bad edges all have cut-distance four, but their support union has only $2t + 2$ edges. Thus Hall expansion fails for $t \geq 3$, with an unbounded ratio. An explicit packing of t^2 edge-disjoint 7-cycles certifies the maximum cut and its uniqueness. We also derive the finite footprint forced by a minimal Hall violation. Exact enumeration gives no footprint with at most eight atoms, one footprint with nine atoms, and three footprint isomorphism classes with ten atoms. The graph G_3 realizes the unique nine-atom obstruction. The family theorem is combinatorial; the finite classification is computer-assisted and has two independent exact verifiers. These results do not resolve the Erdős $n^2/25$ conjecture.

1 Introduction

For a finite graph G , let

$$\text{bip}(G) = |E(G)| - \text{MaxCut}(G)$$

be its edge-deletion distance to bipartiteness. Erdős, Faudree, Pach and Spencer asked whether every triangle-free N -vertex graph satisfies

$$\text{bip}(G) \leq N^2/25. \tag{1}$$

The balanced blow-up of C_5 shows that the constant would be sharp. See [3, 4, 1] for the problem and the strongest general bounds. The conjectured value is known for $N = 5n$, $1 \leq n \leq 40$, by an exact certificate [5].

We study a local packing principle suggested by a fixed maximum cut. Let B be the crossing subgraph and M the edges left monochromatic. For $xy \in M$, the endpoints lie on the same shore. If B is connected, then $d_B(x, y)$ is even; triangle-freeness excludes distance two. The first case is therefore $d_B(x, y) = 4$. One may try to assign a distinct shortest-path edge to every such bad edge by Hall's theorem.

Our main theorem says that this principle fails by an arbitrarily large factor even under unique maximum-cut and unique-geodesic hypotheses. The use of shortest odd cycles in max-cut estimates has precedents, notably [2], but the full shortest-geodesic support incidence system considered here appears to be different.

2 Supports and minimal violations

Fix a cut $V(G) = V_0 \dot{\cup} V_1$, and set

$$B = \{xy \in E(G) : x \in V_0, y \in V_1\}, \quad M = E(G) \setminus B.$$

Assume that B is connected.

Definition 2.1. For $e = xy \in M$, define

$$\text{supp}_B(e) = \{f \in B : f \text{ lies on a shortest } x\text{-}y \text{ path in } B\}.$$

For $\mathcal{S} \subseteq M$, let $\text{supp}_B(\mathcal{S}) = \bigcup_{e \in \mathcal{S}} \text{supp}_B(e)$. The length-five bad edges have *support Hall expansion* when

$$|\text{supp}_B(\mathcal{S})| \geq |\mathcal{S}| \tag{2}$$

for every $\mathcal{S} \subseteq \{xy \in M : d_B(x, y) = 4\}$.

Condition (2) is Hall's condition for the incidence graph between bad edges and the crossing edges in their shortest supports.

Proposition 2.2 (Support-size gap). *If B is bipartite and $d_B(x, y) = 4$, then*

$$|\text{supp}_B(xy)| = 4 \quad \text{or} \quad |\text{supp}_B(xy)| \geq 6.$$

Proof. If the shortest path is unique, its four edges form the support. Otherwise choose two distinct shortest paths P, Q . Their nonempty symmetric difference is an even-degree subgraph and contains a cycle. Since B is bipartite, this cycle has at least four edges. Hence

$$|E(P) \cup E(Q)| = \frac{4 + 4 + |E(P) \triangle E(Q)|}{2} \geq 6.$$

□

Lemma 2.3 (Minimal footprint). *Let G be triangle-free, and let $\mathcal{S}$ be inclusion-minimal with*

$$|\text{supp}_B(\mathcal{S})| < |\mathcal{S}| = m.$$

Writing $F = \text{supp}_B(\mathcal{S})$, we have:

- (i) *F is connected and bipartite, with $|E(F)| = m - 1$;*
- (ii) *every edge of F belongs to at least two atom supports;*
- (iii) *every $xy \in \mathcal{S}$ has distance four in F , and its support is the union of all length-four F -geodesics from x to y ;*
- (iv) *the graph $(V(F), \mathcal{S})$ is triangle-free.*

Proof. For every $e \in \mathcal{S}$, minimality gives

$$|\text{supp}_B(\mathcal{S} \setminus \{e\})| \geq m - 1.$$

This set is contained in F , while $|E(F)| < m$. Hence $|E(F)| = m - 1$ and removing any atom leaves the whole support union. This proves (ii).

Every atom support is connected. If F had several components, the atoms partitioned among them, and one component would itself contain more atoms than support edges, contradicting minimality. Thus F is connected. It is bipartite because $F \subseteq B$.

A shortest B -path for an atom lies in F , so its F -distance is at most four. The distance is even, and distance two would make a triangle with the bad edge. Thus the distance is four. Every length-four path in F is also a shortest path in B , proving (iii). Finally, $\mathcal{S} \subseteq E(G)$, proving (iv). □

3 An infinite Hall obstruction

Let L, A, B, C, D, E, R be seven disjoint independent sets of size t , and let u, w, v be three further vertices. Define G_t by adding all edges between consecutive classes in

$$L - A - B - C - D - E - R,$$

all edges of $L - R$, and the thin channel

$$L - u - w - v - R,$$

where $L - u$ and $v - R$ are complete blocks. Then

$$|V(G_t)| = 7t + 3, \quad |E(G_t)| = 7t^2 + 2t + 2.$$

The graph is triangle-free: neither channel has a chord, and no vertex outside $L \cup R$ is adjacent to both terminal classes.

Theorem 3.1. *For every $t \geq 1$, G_t has a unique maximum cut up to global complementation, and*

$$\text{bip}(G_t) = t^2, \quad \text{MaxCut}(G_t) = 6t^2 + 2t + 2.$$

The bad edges are exactly $L - R$.

Proof. Index each large class by $\mathbb{Z}_t$. For every $(i, j) \in \mathbb{Z}_t^2$, form the 7-cycle

$$L_i A_j B_{i+j} C_i D_j E_{i+j} R_j L_i. \quad (3)$$

These t^2 cycles are edge-disjoint. On the six wide-channel blocks, their edge labels are

$$(i, j), (j, i + j), (i + j, i), (i, j), (j, i + j), (i + j, j),$$

and each map is a bijection of $\mathbb{Z}_t^2$. Their closing $L_i R_j$ edges are also distinct.

Every cut leaves a monochromatic edge on each packed odd cycle, so it has at least t^2 bad edges. Conversely, put L, B, D, R, w on one shore and A, C, E, u, v on the other. Exactly the $L - R$ edges are bad.

For uniqueness, a maximum cut has exactly one bad edge in each packed cycle and no bad edge outside the packing. The outside edges are the thin channel. Since all $L_i u, uw, wv, v R_j$ cross, all vertices in $L \cup R$ lie on one shore and u, v on the other. Thus all t^2 edges of $L - R$ are bad, and every remaining edge crosses. This determines the cut up to complementation. $\square$

Corollary 3.2. *In the unique maximum cut of G_t , every bad edge $L_i R_j$ has the unique shortest B -geodesic*

$$L_i - u - w - v - R_j,$$

and

$$|\text{supp}_B(M)| = 2t + 2.$$

Thus (2) fails for $t \geq 3$, and

$$\frac{|M|}{|\text{supp}_B(M)|} = \frac{t^2}{2t + 2} \rightarrow \infty.$$

Proof. After deleting $L - R$, the thin channel has length four and the wide channel has length six. The thin internal vertices are singletons, so the length-four path is unique. The support union consists of the t edges $L - u$, the two edges uw, wv , and the t edges $v - R$. $\square$

The maximum cut is unique and therefore automatically minimizes every secondary tie-break among maximum cuts. In particular,

$$\Gamma(B, M) = \sum_{xy \in M} (d_B(x, y) + 1)^2 = 25t^2.$$

The graphs G_t are not counterexamples to (1), since $t^2 < (7t + 3)^2/25$.

4 Exact classification of the first footprints

Definition 4.1. An m -footprint obstruction is a pair $(F, \mathcal{A})$ such that:

- (a) F is connected and bipartite with $m - 1$ edges;
- (b) $\mathcal{A}$ is a set of m distinct distance-four pairs in F ;
- (c) the graph $(V(F), \mathcal{A})$ is triangle-free;
- (d) the geodesic supports $P_F(a)$ cover $E(F)$, and every edge of F lies in at least two supports.

Lemma 2.3 sends every graph-level minimal Hall violation to an m -footprint obstruction.

Theorem 4.2 (Computer-assisted). *No m -footprint obstruction exists for $m \leq 8$. For $m = 9$, there is exactly one footprint isomorphism class and its atom set is forced:*

$$\{l_1, l_2, l_3\} - u - w - v - \{r_1, r_2, r_3\}, \quad \mathcal{A} = \{l_i r_j : 1 \leq i, j \leq 3\}. \quad (4)$$

For $m = 10$, there are exactly three footprint isomorphism classes.

The cases $m \leq 6$ have a short direct proof. A minimal violation has $m - 1$ support edges. Every atom support has at least four edges, so $m \geq 5$. At $m = 5$, the footprint is a four-edge path and determines only one distance-four pair. At $m = 6$, Proposition 2.2 forces each support to be one of the at most $\binom{5}{4} = 5$ four-edge subsets of a five-edge footprint; a path edge set determines its endpoint pair.

For $7 \leq m \leq 10$, the proof is exhaustive. Using **geng** from nauty/Traces [6], we generated one representative of every connected bipartite graph with $m - 1$ edges and between 5 and m vertices. Breadth-first search listed every distance-four pair and its full geodesic support. We then enumerated every m -element set of pairs and tested conditions (c) and (d). Table 1 gives the exact output. A second, independently written analysis program reproduced the table.

m	distance-filtered footprints	valid footprint classes	admissible atom subsets
6	0	0	0
7	0	0	0
8	0	0	0
9	1	1	1
10	9	3	56

Table 1: Exact footprint enumeration. The final column counts subsets on canonical footprint representatives, before quotienting those subsets by footprint automorphisms.

The graph6 label in (4) is H???FaM. The ten-atom footprint labels are I????Bfo, I????Bobo, and I????E?wh_. Consequently, every graph-level minimal violation has at least nine atoms, and the nine-atom footprint is uniquely determined. The graph G_3 realizes it. More generally, choosing three vertices from each of L and R in any G_t , $t \geq 3$, yields the same minimal subsystem.

5 Interpretation and reproducibility

The wide channel in G_t certifies maximum-cut optimality by packing odd cycles, while the thin channel forces all bad edges to share the same two central support edges. Thus the resources certifying cut optimality and the resources offered by shortest supports can live on different channels.

This rules out bare shortest-support Hall expansion, not stronger banked inequalities that also use vertex slack, longer geodesics, or cut-boundary capacity. It also gives no stronger numerical upper bound on $\text{bip}(G)$.

The ancillary archive contains two family verifiers, two footprint enumerators, a command-level README, and a SHA-256 manifest. The family theorem is independent of computation. The classification uses integer arithmetic throughout and relies on nauty/Traces only for complete unlabeled graph generation.

References

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